SEPM LAB

PRACTICAL -8

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Section : B4

Batch: B2

1. Aim

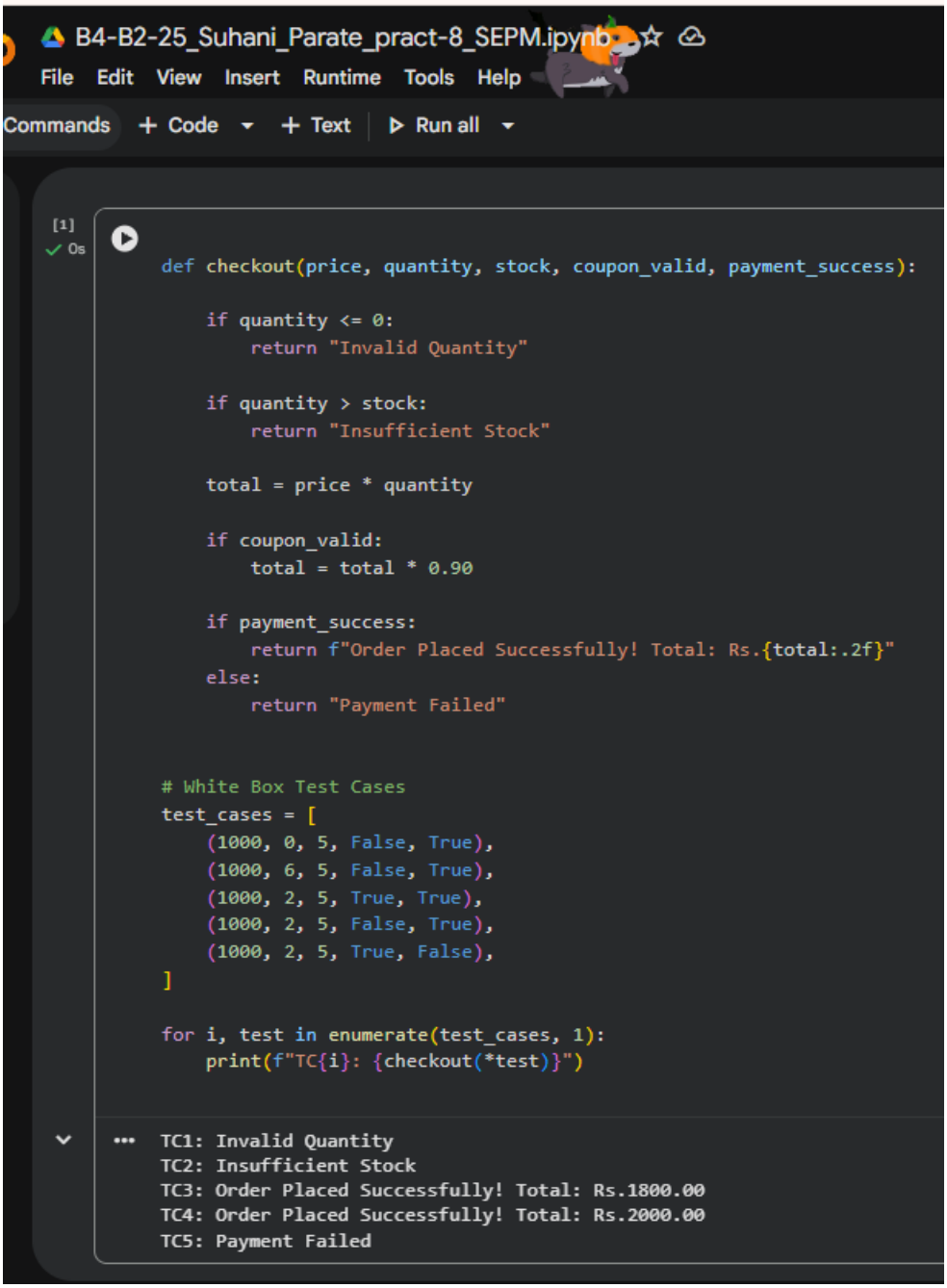
To implement white box testing for the Digital Fashion Shopping System using a Control Flow Graph and cyclomatic complexity to identify independent execution paths and design appropriate test cases.

2. Case Study

A customer selects a fashion product, specifies the quantity, applies an optional discount coupon, and proceeds to payment.

The checkout module verifies whether the requested quantity is valid, checks inventory availability, validates the coupon, and processes payment. If any validation fails, the system displays the appropriate error message.

3. Code


B4-B2-25_Suhani_Parate_pract-8_SEPM.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
[1]
0s
def checkout(price, quantity, stock, coupon_valid, payment_success):
if quantity <= 0:
return "Invalid Quantity"
if quantity > stock:
return "Insufficient Stock"
total = price * quantity
if coupon_valid:
total = total * 0.90
if payment_success:
return f"Order Placed Successfully! Total: Rs.{total:.2f}"
else:
return "Payment Failed"
White Box Test Cases
test_cases = [
(1000, 0, 5, False, True),
(1000, 6, 5, False, True),
(1000, 2, 5, True, True),
(1000, 2, 5, False, True),
(1000, 2, 5, True, False),
]
for i, test in enumerate(test_cases, 1):
print(f"TC{i}: {checkout(*test)}")
TC1: Invalid Quantity
TC2: Insufficient Stock
TC3: Order Placed Successfully! Total: Rs.1800.00
TC4: Order Placed Successfully! Total: Rs.2000.00
TC5: Payment Failed

4. Control Flow Graph Code



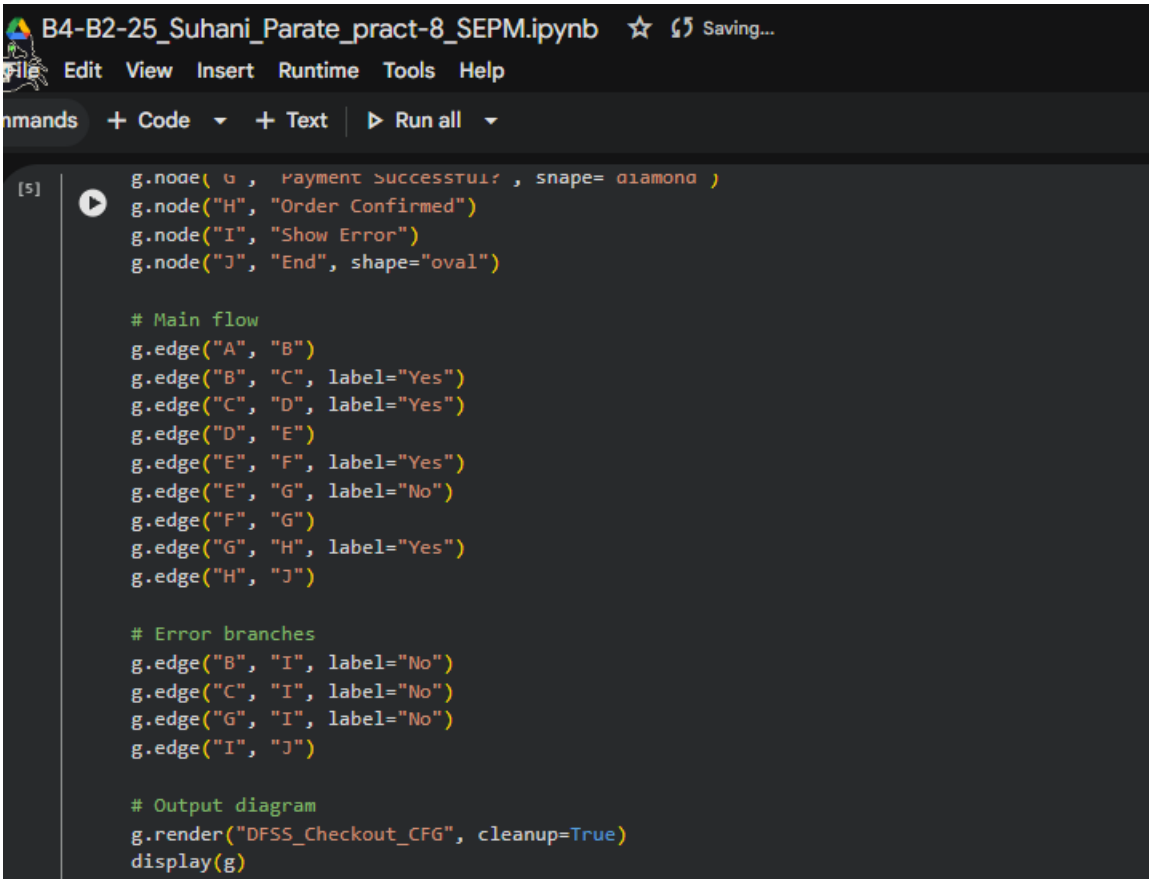
```
!apt-get -qq install graphviz
!pip -q install graphviz

from graphviz import Digraph

g = Digraph("Checkout_CFG", format="png")
g.attr(rankdir="TB", splines="ortho", nodesep="0.45",
       ranksep="0.55", bgcolor="white")
g.attr("node", shape="box", style="solid",
       color="black", fontcolor="black",
       fontname="Arial", fontsize="11", margin="0.15")
g.attr("edge", color="black", fontname="Arial", fontsize="10",
       arrowsize="0.7")

# Nodes
g.node("A", "Start", shape="oval")
g.node("B", "Quantity > 0?", shape="diamond")
g.node("C", "Stock Available?", shape="diamond")
g.node("D", "Calculate Total")
g.node("E", "Coupon Valid?", shape="diamond")
g.node("F", "Apply Discount")
g.node("G", "Payment Successful?", shape="diamond")
g.node("H", "Order Confirmed")
g.node("I", "Show Error")
g.node("J", "End", shape="oval")

# Main flow
g.edge("A", "B")
g.edge("B", "C", label="Yes")
g.edge("C", "D", label="Yes")
g.edge("D", "E")
g.edge("E", "F", label="Yes")
g.edge("E", "G", label="No")
```


B4-B2-25_Suhani_Parate_pract-8_SEPM.ipynb
Saving...
File Edit View Insert Runtime Tools Help
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[5]
g.node(G , Payment Successful? , shape= diamond)
g.node("H", "Order Confirmed")
g.node("I", "Show Error")
g.node("J", "End", shape="oval")
Main flow
g.edge("A", "B")
g.edge("B", "C", label="Yes")
g.edge("C", "D", label="Yes")
g.edge("D", "E")
g.edge("E", "F", label="Yes")
g.edge("E", "G", label="No")
g.edge("F", "G")
g.edge("G", "H", label="Yes")
g.edge("H", "J")
Error branches
g.edge("B", "I", label="No")
g.edge("C", "I", label="No")
g.edge("G", "I", label="No")
g.edge("I", "J")
Output diagram
g.render("DFSS_Checkout_CFG", cleanup=True)
display(g)

Output :

```
g.edge("B", "I", label="No")
g.edge("C", "I", label="No")
g.edge("G", "I", label="No")
g.edge("I", "J")

# Output diagram
g.render("DFSS_Checkout_CFG", cleanup=True)
display(g)
```

